

PULSE Instrument Cluster

Functional Safety Concept, preliminary (ISO 26262)

Ahmed Abdelghany

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Document control

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Change log

Version	Date	Author	Change
0.9	2026-09-03	A. Abdelghany	First skeleton: item definition, safety-relevant signals with freshness proposals, preliminary HARA, safety mechanisms including the minimum-risk manoeuvre, interference argument outline, qualification gap list, SR-1 to SR-11 with status
0.9.1	2026-09-03	A. Abdelghany	Safety-mechanism placement and manoeuvre state-machine diagrams added
0.9.2	2026-09-03	A. Abdelghany	SR-9 evidence updated for the regression suite
0.9.3	2026-09-08	A. Abdelghany	Worked ASIL derivation added for H-7 so the S, E and C columns can be read without knowing the method. SM-2 and the item-definition context reworded. SR-1 evidence tightened. Glossary pointer added
0.9.4	2026-09-23	A. Abdelghany	SM-1 implemented in both HMIs; SR-4, SR-5 and SR-9 to Implemented. Freshness judged by arrival time rather than broker timestamp, with the reason recorded in section 3
0.9.5	2026-10-02	A. Abdelghany	SR-4 and SR-5 verified on the Compose cluster on a device: all eight freshness criteria, instrumented smoke test, captures of the degraded and recovered display. Flutter recovery after a broker restart fixed and recorded on the strip; empty subscription entries no longer count as arrivals

1. Purpose and disclaimer

The PULSE demonstrator is not a qualified safety item and no ASIL is claimed. This document establishes compliance readiness: the artefacts and design decisions a production programme would need, produced as a by-product of the build rather than retrofitted later.

Every rating in the hazard analysis is a proposal by the project engineer and has not been reviewed by a safety manager. Ratings are recorded so that a reviewer has something concrete to disagree with.

Abbreviations used here are defined in the glossary, PULSE-StRS-001 section 8.

2. Item definition (SR-1)

Field	Content
Item name	PULSE cluster and driver-monitoring function
Purpose	Present vehicle state to the driver on a strip display; detect driver drowsiness and distraction from a driver-facing camera; on sustained drowsiness, trigger a minimum-risk manoeuvre
Functions	F1 display of road speed, engine speed, gear, fuel, range, tyre pressures, indicators. F2 display of motorsport telemetry. F3 driver-state estimation and alerting. F4 minimum-risk manoeuvre trigger and status display
System boundary	As PULSE-SAD-001 section 3.1: catalogue, broker, providers, both HMIs, orchestration
Assumed vehicle context	A passenger car or race car with a single driver. The cluster is the primary source of speed and gear information. On a real vehicle a separate motion controller executes braking; on the desk rig the simulator stands in for it
Interfaces crossing the boundary	Camera frames in (V4L2); display surface out (DisplayPort); vehicle bus in (SocketCAN, planned); autonomy stack in (planned); Android surface out
Operational situations	Highway cruise, urban driving, track driving, standstill; day and night (night is backlog for the camera)
Assumptions on other elements	The vehicle bus delivers correct signals. The motion controller executes a commanded stop safely. The display panel renders what it is sent

3. Safety-relevant signals (SR-3, SR-5)

Signals whose corruption or loss could mislead the driver. Each carries a proposed freshness criterion: a value older than the criterion is invalid and shall be shown as degraded, not as its last value.

Signal	Why safety-relevant	Publish rate	Proposed freshness	Degraded presentation
Vehicle.Speed	Speed misjudgement; legal and collision consequences	10 Hz	500 ms	Numerals replaced by dashes, dial greyed
Vehicle.Powertrain.Transmission.SelectedGear	Unintended gear change of travel at low speed	10 Hz	500 ms	Gear glyph replaced by “?”
Vehicle.Powertrain.Combustion.Engine.Speed	Over-revving; secondary damage	10 Hz	500 ms	Dial greyed
Vehicle.Body.Lights.Hazard	Driver may assume the vehicle is signalling a stop	10 Hz on change	2 s	Indicator shown as unknown
Vehicle.Driver.Monitoring.IsDriverMonitoring	Driver may assume monitoring when there is none	10 Hz	1 s	“Monitoring unavailable” banner
Vehicle.Driver.Monitoring.AlertState	Missed alert is a false negative; false alert triggers the manoeuvre	10 Hz	1 s	Treated as NO_FACE (monitoring lost), never as NONE
Vehicle.Driver.Monitoring.InterventionState	Driver may assume the vehicle is about to brake	10 Hz	500 ms	Banner “vehicle state unknown”
Vehicle.Driver.Monitoring.InterventionCountdown	As above	10 Hz	500 ms	Countdown hidden, banner stays

Both HMIs enforce these criteria (SM-1). Freshness is judged by when a value last arrived, on the HMI’s own monotonic clock, rather than by the broker’s timestamp: that needs no clock agreement between hosts and catches a lost broker or link as well as a stalled provider. Providers therefore republish every signal in this table on every tick even when its value is unchanged.

4. Preliminary hazard analysis and risk assessment (SR-2)

Ratings follow ISO 26262-3: severity S0 to S3, exposure E0 to E4, controllability C0 to C3. The ASIL column is what the standard’s table returns for that combination. It is a lookup, not a claim about the software.

Worked example, reading H-7 below, the highest rating in the table. The malfunction is a false detection of drowsiness that starts the manoeuvre while the driver is attentive. The effect is unintended braking in moving traffic, which can injure severely, so severity is S3. It applies during highway cruise, which almost every drive includes, so exposure is E4. A driver who is alert can react and override, so controllability is C2, not C3. S3 with E4 and C2 returns ASIL C. Every other row reads the same way.

ID	Malfunction	Operational situation	Effect	S	E	C	ASIL proposal	Safety goal
H-1	Speed displayed too low	Any driving	Driver exceeds intended speed	S1	E4	C2	A	SG-1 The displayed speed shall not be lower than the actual speed by more than a defined tolerance

ID	Malfunction	Operational situation	Effect	S	E	C	ASIL pro-posal	Safety goal
H-2	Speed display frozen at last value	Any driving	Driver trusts a stale value; worse than blank	S2	E4	C2	B	SG-2 A stale speed shall be recognisable as stale within its freshness criterion
H-3	Wrong gear indicated (D shown, R engaged)	Manoeuvring at standstill	Vehicle moves in an unexpected direction	S2	E3	C2	A	SG-3 Gear indication shall match the engaged gear or show unknown
H-4	Display blank	Any driving	Loss of information, driver aware	S1	E4	C1	QM	SG-4 Loss of display shall be obvious, not plausible
H-5	Hazard or stop indication missing during the manoeuvre	Highway cruise, manoeuvre active	Driver surprised by braking	S2	E2	C2	QM	SG-5 Manoeuvre state shall be displayed before actuation begins
H-6	Drowsiness not detected (false negative)	Highway cruise, drowsy driver	No alert, no manoeuvre	S3	E2	C2	A	SG-6 Sustained drowsiness shall raise an alert within the hold time
H-7	Drowsiness falsely detected, manoeuvre executed	Highway cruise, attentive driver	Unintended braking in traffic	S3	E4	C2	C	SG-7 The manoeuvre shall not begin without a driver-recovery window and a visible countdown
H-8	Alert shown but manoeuvre state stale	Manoeuvre in progress	Driver cannot predict vehicle behaviour	S2	E2	C2	QM	SG-5 as above

H-7 is the finding to carry forward: the actuation side of the loop, not the display, would carry the highest ASIL in a production programme. That is why the manoeuvre is specified as a safety mechanism with a recovery window (SM-3) rather than as a feature.

5. Safety mechanisms

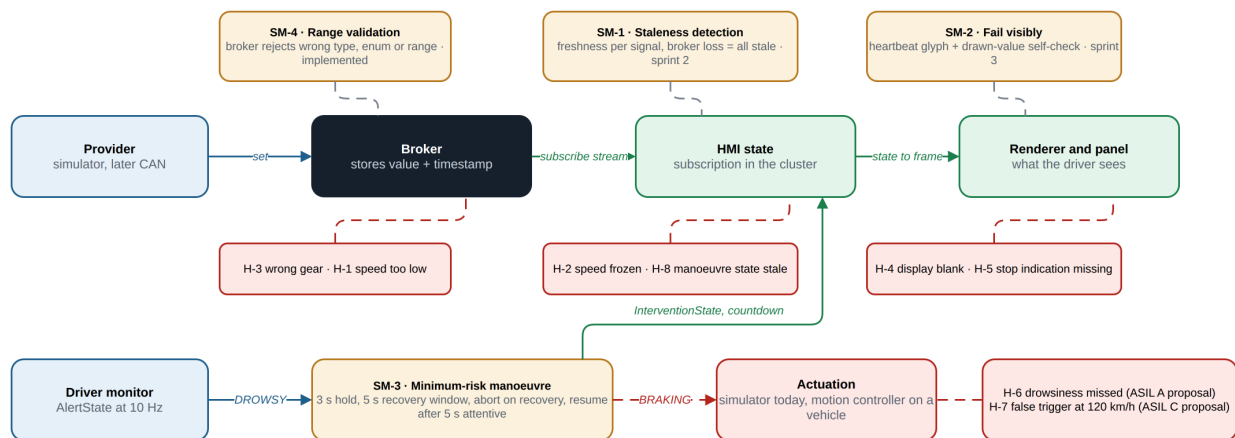


Figure 1: Where each safety mechanism sits on the signal path, and which hazards it addresses. Ratings are proposals.

Source: docs/diagrams/safety-mechanisms.drawio.

SM-1 Staleness detection and degraded display (SR-4, SR-5)

Each HMI shall track the arrival time of every safety-relevant signal and compare it to the freshness criterion in section 3 at least every 100 ms. Loss of the broker connection shall be treated as all signals stale at once. The degraded presentation in section 3 shall replace the value. Implemented in sprint 2: the Flutter cluster degrades within one second of the broker stopping, verified on screen and by widget test; the Compose cluster carries the same model, verified by compilation and unit test, not yet on a device.

SM-2 Fail visibly (SR-6)

A rendering fault shall not present a plausible but wrong value. Two mechanisms are proposed for the demonstrator. First, a heartbeat glyph that the renderer toggles every frame from the signal timestamp, so a frozen renderer becomes visible within one second. Second, a self-check that the speed numeral actually drawn matches the state value. A production programme would need more than this: a qualified renderer, or a supervisor that verifies the output (DD-1 in PULSE-SAD-001). Planned sprint 3.

SM-3 Minimum-risk manoeuvre on sustained drowsiness (SR-7)

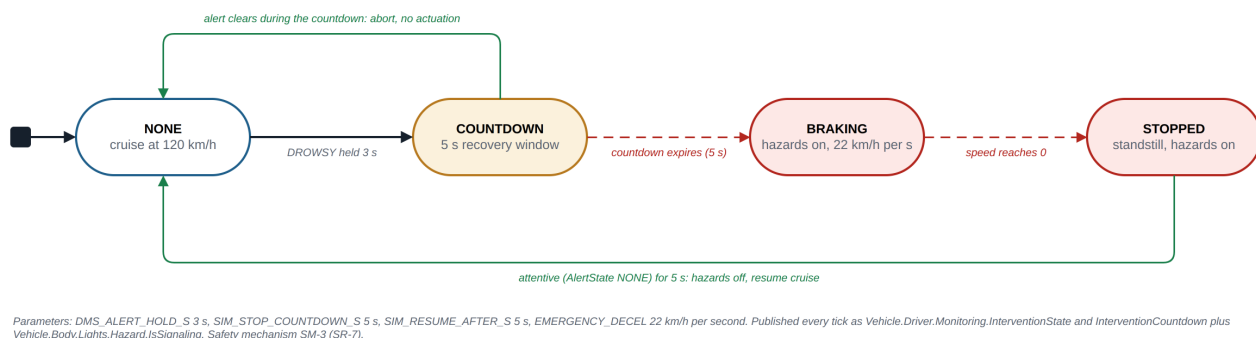


Figure 2: SM-3 as a state machine. Trigger, recovery window, escalation, abort and resume, with the current parameters.

Source: docs/diagrams/mrm-state-machine.drawio.

Element	Specification	Current value
Trigger condition	AlertState DROWSY sustained beyond the alert hold time, while in cruise mode	Hold 3.0 s (DMS_ALERT_HOLD_S), PERCLOS over 30 s window above 0.40, or microsleep over 1.5 s
Driver-recovery window	A countdown displayed on the cluster during which the alert clearing aborts the manoeuvre	5 s (SIM_STOP_COUNTDOWN_S)
Escalation	Hazard lights on, cruise disengaged, controlled braking to standstill	22 km/h per second, about 6 m/s squared
Abort condition	AlertState leaves DROWSY during the countdown	Immediate return to NONE
Resume	Driver attentive (AlertState NONE) for a sustained period after standstill	5 s (SIM_RESUME_AFTER_S)
Status display	InterventionState and InterventionCountdown published every tick and rendered as a banner	Implemented

Implemented in the simulator's cruise mode (FR-20). On a real vehicle the escalation moves to a motion controller; the trigger and status signal contract stays.

SM-4 Range validation at the broker (shared with CS-5)

The broker rejects values outside the catalogue's datatype, allowed list or range, so a faulty provider cannot drive an impossible value onto the display. Implemented by KUKSA; a client-side check is added in sprint 2.

6. Freedom from interference (SR-8)

Argument outline for the mixed-criticality cockpit, to be completed in sprint 4:

- Spatial: the cluster and any infotainment surface run as separate processes; on the target, separate Wayland surfaces with the cluster surface always on top.
- Temporal: the cluster's render loop does not depend on any infotainment process; broker fan-out is per-stream, so a slow subscriber does not delay another.
- Communication: infotainment is a subscriber only (CS-4 write authorisation); it cannot publish a safety-relevant signal.

- Evidence to produce: a stress test in which an infotainment process saturates CPU and the cluster's frame time is measured.

7. Documented deviations (SR-10)

Held in PULSE-SAD-001 section 8, decisions DD-1 to DD-11. The ones that would block a later ASIL argument are DD-1 (unqualified renderer), DD-3 (container runtime), DD-7 (desk window system) and DD-9 (actuation in the simulator).

8. Qualification gap list (SR-11)

Component	What exists	What a production programme would still need
Linux kernel and board support KUKSA databroker	Ubuntu on x86-64; Raspberry Pi 5 planned Open-source Rust broker	Safety-certified OS or a hypervisor partition for the cluster; qualified BSP Safety argument for the broker as a QM element with monitoring, or a qualified alternative; tool qualification not applicable
VSS toolchain	vss-tools at a pinned version	Tool confidence level assessment for the catalogue compiler
Flutter engine and renderer Compose on Android Automotive Driver monitor	Upstream Flutter, unqualified AOSP emulator MediaPipe model on CPU	Qualified renderer or output-verifying supervisor (SM-2) Not a candidate for the safety-relevant surface; infotainment only Evaluated dataset with known false-positive and false-negative rates; infrared camera; per-driver calibration; SOTIF (ISO 21448) analysis for the perception function
Simulator and actuation	Python simulator	Real motion controller with its own safety concept; PULSE only provides the trigger and status signals
Toolchain	Dart, Kotlin, Python, Bash	Qualified compilers where the safety-relevant code runs; coding guideline compliance

9. Safety requirements

ID	Requirement	Verification	Status (W3)	Evidence
SR-1	An item definition shall be written for the cluster and driver-monitoring function, naming the system boundary, the assumed vehicle context, and the interfaces crossing that boundary (ISO 26262-3).	I	Partial	Section 2 of this document
SR-2	A preliminary hazard analysis and risk assessment (HARA) shall be recorded for the displayed functions, with severity, exposure and controllability rated per malfunction and a resulting ASIL proposal.	I	Partial	Section 4; eight hazards rated; review in S2

ID	Requirement	Verification	Status (W3)	Evidence
SR-3	Signals whose corruption or loss could mislead the driver (road speed, gear, warning indicators, drowsiness alert) shall be identified as safety-relevant and listed explicitly.	I	Implemented	Section 3
SR-4	The system shall detect loss of a safety-relevant signal, including broker disconnection and stale values, and shall indicate the degraded state rather than continuing to display the last known value.	T	Implemented	SM-1. Flutter: pulse-cluster/test/cluster_smoke_test.dart, arrival_test.dart, and captures on the physical strip live, one second after the broker was stopped, and recovered within 4 s of its return (docs/evidence/screens/20261002-flutter-strip-*). Compose: FreshnessTest.kt, the instrumented ClusterSmokeTest.kt on the Android 16 car emulator (suite check compose-ui), and captures 1.4 s after the broker was stopped and after it returned, with no action on the app (docs/evidence/screens/20261002-compose-*)
SR-5	Every safety-relevant signal shall carry a freshness criterion; a value older than its criterion shall be treated as invalid.	T	Implemented	Criteria in section 3 are the kFreshness table in vehicle_state.dart and FRESHNESS_NS in VssClient.kt, all eight in both since 2 October; providers republish them every tick (KEEPALIVE)
SR-6	Safety-relevant display elements shall fail visibly. A rendering fault shall not present a plausible but wrong value.	T	Planned S3	SM-2
SR-7	The minimum-risk manoeuvre triggered by sustained driver drowsiness shall be specified as a safety mechanism: trigger condition, driver-recovery window, escalation, and abort condition.	D	Implemented	SM-3; implemented in emulator/telemetry_sim.py cruise mode; demonstrated live
SR-8	Freedom from interference shall be argued for the mixed-criticality cockpit: infotainment content shall not be able to degrade or obscure the safety-relevant cluster surface.	A	Planned S4	Section 6 outline

ID	Requirement	Verification	Status (W3)	Evidence
SR-9	A safety-relevant requirement shall be traceable to the verification that demonstrates it, and each verification result shall be reproducible from the repository.	T	Implemented	PULSE-RTM-001 generated from this table; 16-check suite runs before each demo and in CI, report committed to docs/evidence/; SR-4 and SR-5 have named tests
SR-10	Any decision that would block a later ASIL argument, such as the choice of graphics stack, operating system or toolkit, shall be recorded as a documented deviation with its rationale.	I	Implemented	PULSE-SAD-001 section 8, DD-1 to DD-11
SR-11	A qualification gap list shall be maintained: for each component, what a production programme would still need (qualified toolchain, certified OS, evidence of a safety-qualified renderer).	I	Implemented	Section 8

10. Verification and traceability

Safety requirements are traced to user stories US-13, US-15 and US-17 in PULSE-StRS-001 and to their verification in PULSE-RTM-001. Test cases for SR-4, SR-5 and SR-6 are written in sprint 2 as part of the regression suite so that each demo re-runs them.